THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The number of factors of 36 is: 1

(a) 8 **(b)** 10 **(c)** 9 **(d)** 11

2. 60 mL of a 15% sugar solution is mixed with 120 mL of a 33% sugar solution. The concentration of the mixture is: 1

(a) 29% **(b)** 27% **(c)** 28% **(d)** 26%

3. The value of $146 \bmod 3$ (remainder on dividing 146 by 3) is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 3

4. The diagonals of a parallelogram intersect at O. If $OA = (2x + 8)$ cm and $x = 3$, then the diagonal AC is: 1

(a) 28 cm **(b)** 30 cm **(c)** 42 cm **(d)** 32 cm

5. The slope of the line $x - y = 6$ is: 1

(a) 3 **(b)** 2 **(c)** 0 **(d)** 1

6. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 10.5 cm and height 14 cm is: 1

(a) 1779 cm^3 **(b)** 1617 cm^3 **(c)** 1942 cm^3 **(d)** 1455 cm^3

7. If $x < 0$ and $y > 0$, the point (x, y) lies in: 1

(a) Quadrant III **(b)** Quadrant IV **(c)** Quadrant II **(d)** Quadrant I

8. The graph of $y = 2x - 8$ cuts the x-axis at: 1

(a) $(4, 0)$ **(b)** $(5, 0)$ **(c)** $(0, 4)$ **(d)** $(-4, 0)$

9. A rational number $\frac{p}{q}$ equal to $0.1\overline{6}$ is: 1

(a) 0 **(b)** $\frac{1}{7}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{3}$

10. $x^2 + 8x + 16$ is the square of: 1

(a) $(x + 4)$ **(b)** $(x + 8)$ **(c)** $(x + 16)$ **(d)** $(x - 4)$

11. In a cyclic quadrilateral $ABCD$, $\angle A = 55^\circ$ and the side BC is produced to E. Then $\angle DCE =$ 1

(a) 75° **(b)** 65° **(c)** 45° **(d)** 55°

12. The area of a rhombus with diagonals 14 cm and 10 cm is (cm^2): 1

(a) 71 **(b)** 70 **(c)** 69 **(d)** 72

13. A bag has 4 red, 3 black and 5 white balls. The probability of drawing a ball that is not black is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{1}{4}$

14. The sum of the first 10 terms of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 230 **(b)** 210 **(c)** 200 **(d)** 220

15. A triangle has sides 9 cm, 40 cm and 41 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) equilateral **(b)** not right-angled **(c)** isosceles **(d)** right-angled

16. In a 100% stacked bar of a household's monthly water use, appliances take 17% of 900 units. The units used by the appliances are: 1

(a) 155 **(b)** 152 **(c)** 153 **(d)** 154

17. In the divisor algorithm for n, the numbers j that are checked are: 1

(a) $1, 2, \ldots, n$ **(b)** only the primes up to n

(c) only the even numbers up to n **(d)** $2, 3, \ldots, n-1$ only

18. If one pair of opposite sides of a quadrilateral is equal and parallel, then it is a: 1

(a) rhombus always **(b)** trapezium only **(c)** kite **(d)** parallelogram

19. If $(5, 2)$ is a solution of $kx + y = 42$, then k is: 1

(a) 8 **(b)** 7 **(c)** 10 **(d)** 9

20. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 35 cm is: 1

(a) 18482 cm^2 **(b)** 16941 cm^2 **(c)** 13859 cm^2 **(d)** 15400 cm^2

21. The distance of $(5, -7)$ from the x-axis is: 1

(a) 6 **(b)** 8 **(c)** 9 **(d)** 7

22. The sum of the zeros of $2x - 6$ and $5x - 20$ is: 1

(a) 9 **(b)** 6 **(c)** 7 **(d)** 8

23. $(-15) - (-23) =$ 1

(a) 7 **(b)** 10 **(c)** 8 **(d)** 9

24. Factorise $x^2 + 7x + 10$: 1

(a) $(x + 4)(x + 5)$ **(b)** $(x + 2)(x + 5)$ **(c)** $(x - 2)(x - 5)$ **(d)** $(x + 3)(x + 4)$

25. In triangle OAB, $OA = OB$ and $\angle AOB = 80^\circ$ (O is the centre). Then $\angle OAB =$ 1

(a) 40° **(b)** 70° **(c)** 50° **(d)** 60°

26. The area of a cyclic quadrilateral with sides 4 cm, 5 cm, 7 cm and 10 cm is (cm^2): 1

(a) 38 **(b)** 36 **(c)** 35 **(d)** 37

27. If $P(E) = 0.3$, then $P(\text{not } E) \times 100$ equals: 1

(a) 69 **(b)** 71 **(c)** 70 **(d)** 72

28. The 5th term of the GP $2, 6, 18, \ldots$ is: 1

(a) 162 **(b)** 164 **(c)** 160 **(d)** 163

29. For the expression $n^2 + n + 11$, the smallest natural number n for which the value is NOT prime is: 1

(a) 9 **(b)** 12 **(c)** 11 **(d)** 10

30. The weighted average of values x_1, x_2 with weights w_1, w_2 is: 1

(a) $\dfrac{w_1x_1 + w_2x_2}{w_1 + w_2}$ **(b)** $\dfrac{w_1 + w_2}{x_1 + x_2}$ **(c)** $w_1x_1 + w_2x_2$ **(d)** $\dfrac{x_1 + x_2}{w_1 + w_2}$

31. Using Euclid's subtraction algorithm for gcd(90, 50), the number of subtraction steps needed is: 1

(a) 5 **(b)** 8 **(c)** 7 **(d)** 6

32. In a rhombus ABCD, $\angle A = 50^\circ$. The angle between the diagonal AC and the side AB is: 1

(a) 25° **(b)** 90° **(c)** 75° **(d)** 35°

33. Which of the following points lies on the line $8x + 9y = 12$? 1

(a) $(-4, 6)$ **(b)** $(6, -4)$ **(c)** $(7, -4)$ **(d)** $(6, -3)$

34. Take $\pi = \frac{22}{7}$. The volume of a sphere of radius 31.5 cm is: 1

(a) 144075 cm^3 **(b)** 130977 cm^3 **(c)** 117879 cm^3 **(d)** 157174 cm^3

35. If the midpoint of AB is $(5, 7)$ and $A = (2, 3)$, then B is: 1

(a) $(8, 11)$ **(b)** $(-8, 11)$ **(c)** $(8, -11)$ **(d)** $(11, 8)$

36. The nth term of the pattern 1, 7, 13, 19, ... is: 1

(a) $6n - 4$ **(b)** $6n - 5$ **(c)** $7n - 5$ **(d)** $6n - 6$

37. Which of the following has a terminating decimal expansion? 1

(a) $\frac{7}{16}$ **(b)** $\frac{2}{7}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{6}$

38. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2 =$ 1

(a) 52 **(b)** 54 **(c)** 53 **(d)** 55

39. Equal chords of a circle are ___ from the centre. 1

(a) at distances in ratio $1 : 2$ **(b)** at different distances

(c) on the same line **(d)** equidistant

40. The length of an arc of angle 120° in a circle of radius 21 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 46 **(b)** 44 **(c)** 45 **(d)** 43

41. A fair die is thrown once. The probability of getting a composite number is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{6}$

42. The number of terms in the AP $3, 7, 11, 15, \ldots$ ending at 83 is: 1

(a) 21 **(b)** 22 **(c)** 20 **(d)** 23

43. A proposition that is false can be shown to be false by giving: 1

(a) its converse **(b)** many examples that satisfy it

(c) a definition **(d)** a counterexample

44. The average of 30 numbers is 54. The average of all 45 numbers (including 15 more numbers) is 46. The average of the 15 additional numbers is: 1

(a) 29 **(b)** 31 **(c)** 30 **(d)** 32

45. Set x = 8. Repeat 6 times: x = 2x − 1. The final value of x is: 1

(a) 448 **(b)** 449 **(c)** 451 **(d)** 450

46. The sides of a triangle are 34 cm, 38 cm and 14 cm. The perimeter of the triangle formed by joining the midpoints of its sides is: 1

(a) 64 cm **(b)** 45 cm **(c)** 43 cm **(d)** 129 cm

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47. A line with slope -1 passes through $(-1, -2)$. Its y-intercept is: 1

(a) -3 **(b)** -1 **(c)** -2 **(d)** -4

48. Take $\pi = \frac{22}{7}$. The curved surface area of a hemisphere of radius 3.5 cm is: 1

(a) 77 cm^2 **(b)** 85 cm^2 **(c)** 94 cm^2 **(d)** 69 cm^2

49. The point $(-3, 8)$ is reflected in the y-axis. The image is: 1

(a) $(3, 8)$ **(b)** $(8, 3)$ **(c)** $(3, -8)$ **(d)** $(-3, 8)$

50. A plumber charges ₹100 plus ₹50 per hour. For 3 hours he charges ₹: 1

(a) 251 **(b)** 249 **(c)** 250 **(d)** 252

END OF QUESTION PAPER **Total: 50 Marks**

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